

CASE REPORT

Spontaneous remission of Cronkhite–Canada syndrome without immunosuppressive medication in the context of decompensated cirrhosis: A case report

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Key Clinical Message

We present a case of Cronkhite–Canada syndrome in a patient with decompensated cirrhosis who had successful induction of remission with nutritional supplementation alone. We propose that early institution of high-protein, high-energy enteral supplementation should be offered to all patients, especially those with compelling contraindications to immunosuppression.

KEYWORDS

Cronkhite–Canada syndrome, decompensated cirrhosis, hematochezia, hamartomatous polyposis syndromes, venous thrombosis

1 | INTRODUCTION

Cronkhite–Canada syndrome (CCS) is a rare, non-hereditary syndrome characterized by diffuse gastrointestinal polyposis and cutaneous manifestations.^{1–4} CCS typically affects middle-aged men and is most commonly reported in Japan (75% of cases).⁴ The etiology of this syndrome is poorly understood. The presence of inflammatory cell infiltration within colonic polyps, anti-nuclear antibodies, as well as reversibility of the polyps and other symptoms with immunosuppression, suggests that the disease is likely secondary to immune dysregulation.^{2–4} It is suspected that there is a combination of genetic and environmental factors that lead to the development of the disease phenotype, with a recent study proposing three candidates for genes that may be related to CCS

pathogenesis: *USP24*, *KCNQ5*, and *FKBP10*.⁵ *USP24* is a regulator of the tumor suppressor gene p53, which there is some previous evidence to suggest has abnormal function in CCS leading to the development of multiple polyps in the gastrointestinal tract.

Patients with CCS usually follow one of three clinical courses, either the spontaneous remission type, relapsing–remitting type, or the chronic continuous type.³ Symptoms include anorexia, abdominal pain, diarrhea, hematochezia, and weight loss. The cutaneous manifestations of onychodystrophy, hyperpigmentation, and alopecia are characteristic of the syndrome. Other complications include protein-losing enteropathy with subsequent edema, dysgeusia, gastrointestinal bleeding, thrombosis, and recurrent infections.^{2–4} The hallmark feature of CCS is multiple hamartomatous polyps located anywhere from the

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esophagus to the colon. Histologically, these polyps show hyperplastic change, edema, and an inflammatory cell infiltration with positive staining for IgG4 plasma cells. The neoplastic potential of these polyps is controversial, but the overall risk of gastrointestinal malignancy does appear to be increased.^{6,7} Prognosis is difficult to estimate due to the rarity of the disease and heterogeneous course, but CCS is cited as having a 55% 5-year mortality rate, with patients most commonly succumbing to gastrointestinal malignancy, gastrointestinal bleeding, thrombosis, or complications of malnutrition.^{3,6}

Although there are no universally accepted guidelines for management of CCS, immunosuppression is generally considered first-line therapy.^{3,6,7} Oral prednisolone doses of 30–50 mg/day have been shown to be effective in inducing remission in most cases.^{3,4} There are a small number of case reports using steroid-sparing agents including cyclosporine, azathioprine, and TNF α inhibitors to induce remission in refractory cases.^{8,9} Nutritional therapy has been described as an important adjunctive therapy to prevent complications and shorten time to remission, but it has also been described as an alternative treatment for induction of remission.^{3,4,10,11}

Our case describes hematochezia as a presenting complaint leading to the rare diagnosis of CCS in a patient with decompensated cirrhosis. To our knowledge, this is the first case of CCS occurring in a patient with liver cirrhosis that has been published. This case highlights the challenges faced in managing CCS in medically co-morbid patients.

2 | CASE HISTORY

A 55-year-old Caucasian male presented to a rural Australian hospital with a 24-h history of hematochezia. Specifically, he had three episodes of large-volume hematochezia, associated with left lower quadrant abdominal discomfort. Symptoms were preceded by 6 months of diarrhea, subjective loss of muscle mass, and an estimated 20%–25% weight loss.

His past medical history was significant for decompensated cirrhosis secondary to metabolic dysfunction-associated fatty liver disease, diagnosed initially in 2019 with an otherwise unremarkable workup for other causes of chronic liver disease (see Table 1). The diagnosis of cirrhosis was made prior to his current presentation and is not thought to be related to CCS, which is supported by an endoscopy 7 months prior to presentation having no evidence of polyposis. Complications included portal hypertension with esophageal varices, recurrent ascites requiring second weekly large volume paracentesis, hepatic encephalopathy, and recurrent spontaneous bacterial

peritonitis (SBP). His medical history also included stable multifactorial stage IIIa chronic kidney disease without proteinuria, panhypopituitarism following resection of a craniopharyngioma at age 12, metabolic syndrome, recurrent ischemic cerebrovascular accidents with memory impairment, recurrent falls, and multiple non-melanomatous skin cancers. His regular medications were aspirin, cholecalciferol, desmopressin, ferrous fumarate, folic acid, frusemide, lactulose, levothyroxine, norfloxacin, pantoprazole, rifaximin, rosuvastatin, spironolactone, and Vitamin A. He was on hydrocortisone 28 mg daily as part of his replacement therapy for panhypopituitarism. He also received testosterone undecanoate 1000 mg/4 mL injections every 3 months. He was a lifelong non-smoker of cigarettes and had no history of alcohol consumption or illicit drug use. There was no family history of gastrointestinal conditions.

On examination, he had left lower quadrant abdominal tenderness on light palpation without peritonism. A per-rectal examination revealed a small amount of fresh blood, an anal fissure at the 6 o'clock position, and small internal rectal hemorrhoids. He had no signs of hemorrhagic shock and was hypervolemic with large volume ascites and bilateral pitting edema. There were additional stigmata of chronic liver disease, including widespread ecchymoses.

3 | METHODS

Initial laboratory findings (summarized in Table 2) were significant for a stable hemoglobin and platelet count and mildly elevated inflammatory markers. A computed tomography scan of his abdomen and pelvis demonstrated a new superior mesenteric vein (SMV) thrombus not present on comparison imaging from 1 month prior. Features of chronic liver disease and portal hypertension were similar to prior imaging. There was no evidence of lymphadenopathy or solid organ malignancy. A colonoscopy was then performed at the local hospital to determine if there was any evidence of a treatable lower gastrointestinal bleeding source. Unexpectedly, the procedure revealed more than 30 sessile polyps from the caecum to the proximal transverse colon, ranging from 2 to 10 mm in size (Figure 1). There was oozing from polyps at the hepatic flexure, which then worsened after 18 cold snare polypectomies were performed for histology. Histology revealed an identical appearance in all polyps. There was elongation and dilatation of the colonic crypts and expansion of the stroma by a combination of edema and inflammatory cell infiltration, mainly lymphocytes and eosinophils (Figure 2). Occasional crypts contained neutrophils and neutrophil debris. There was no epithelial dysplasia. The

TABLE 1 Liver screen performed at diagnosis of cirrhosis in our patient.

Laboratory test	Result	Units	Normal range
Anti-nuclear antibody	Negative	titre	
Anti-neutrophil cytoplasmic antibody	Negative	titre	
Anti-mitochondrial antibody	Negative	titre	
Anti-smooth muscle antibody	Negative		
Anti-liver/kidney microsome antibody	Negative	titre	
Hepatitis B surface antigen (EIA)	Non-reactive		
Hepatitis B surface antibody	<10	IU/L	
Hepatitis B core total antibody	Non-reactive		
Hepatitis C IgG serology (EIA)	Non-reactive		
CMV IgG (EIA)	Reactive		
EBV IgG (EIA)	Reactive		
HIV antigen/total antibody screen	Non-reactive		
Alpha-1-antitrypsin	1.59	mg/L	0.90–2.00
HbA1c	4.6	%	4.3–6.0
TSH	0.10	mU/L	0.3–4.5
Immunoglobulin G Total	13.5	g/L	7.0–16.0
Immunoglobulin G4	1.21	g/L	0.03–2.01
Copper	15	μmol/L	11–19
Ceruloplasmin	239	mg/L	180–300
Copper/ceruloplasmin Ratio	8.3	mol/mol	7.0–10.0
Ferritin	255	μg/L	45–715
Transferrin Saturation	24	%	15–45
Alpha Fetoprotein	4.1	μg/L	<10
Peak Weight	130	kg	
Alcohol Exposure	Nil		
Herbal and Dietary Supplements	Vitamin A, Folic acid		

polyps were considered to be hamartomatous or inflammatory in nature. Plasma cells were infrequent, and immunohistochemistry for IgG and IgG4 showed no significant increase in IgG4-reactive plasma cells. At this point, given the complexities of the case, he was transferred to the tertiary referral center for ongoing management.

On arrival at the tertiary hospital, he had had no further hematochezia and had a stable hemoglobin. Further examination revealed alopecia, fingernail and toenail onychodystrophy, temporal wasting, and proximal myopathy (Figures 3 and 4). The patient met the new diagnostic criteria developed by Hokari et al.³ for 'definite CCS', by having all 3 major criteria: (a) multiple non-neoplastic polyposis in the gastrointestinal tract, (b) gastrointestinal symptoms, and (c) at least one cutaneous manifestation. At the time of diagnosis by Hokari et al.³'s criteria he had active severe clinical and endoscopic disease.

Additional serological and fecal testing showed an elevated fecal alpha-one antitrypsin to quantitative

hemoglobin ratio on two occasions of 0.44 and 0.36 (≤ 0.03), suggestive of protein-losing enteropathy. 24 h fecal protein loss was unable to be obtained as this test was unavailable in the jurisdiction. Serum IgG3 subclass was elevated at 1.70 g/L (0.11–0.85); otherwise, he had normal total globulin and IgG subclass levels. Lupus anticoagulant was positive, and anti-thrombin III was low at 0.72 U/mL (0.83–1.28). A nutrient screen was within normal limits, although the patient was using oral vitamin D and iron supplements. Additional laboratory testing is summarized in Table 3.

Subsequent upper gastrointestinal endoscopy revealed Grade III esophageal varices, moderate non-bleeding portal hypertensive gastropathy, and multiple non-bleeding sessile hamartomatous polyps in the gastric antrum (Figure 5). A biopsy of one of the antral polyps revealed hyperplastic glands within an inflamed, expanded fibrovascular stroma consistent with an hamartomatous polyp (Figure 2).

TABLE 2 Summary of pertinent laboratory findings performed on admission in our patient.

Laboratory test	Result	Unit	Normal range
Hemoglobin	135	g/L	135–180
MCV	85	fL	80–100
White Cell Count	14.6	$\times 10^9/L$	4.0–11.0
Platelet Count	131	$\times 10^9/L$	140–400
C-Reactive Protein	14	mg/L	<5.0
Urea	10.3	mmol/L	2.1–7.1
Creatinine	140	$\mu\text{mol/L}$	60–110
Urea/creatinine Ratio	79	Ratio	40–100
Bilirubin (Total)	17	$\mu\text{mol/L}$	<20
INR	1.3		0.9–1.2
Albumin	27	g/L	35–50

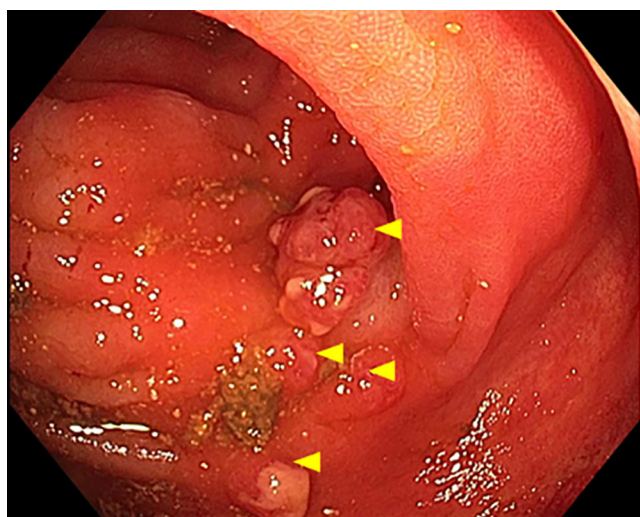


FIGURE 1 Endoscopic images of colonic polyps.

A differential diagnosis of portal hypertensive polyposis was considered given the patient's underlying cirrhosis with portal hypertension. Portal hypertensive polyposis develops mainly in the stomach unlike the pan-gastrointestinal tract distribution typical of CCS. Histologically, portal hypertensive polyps show a triad of mucosal hyperplasia, extensive vascular proliferation, and granulation tissue formation¹² while CCS polyps are less vascular. More importantly, CCS polyps show prominent stromal inflammation and edema.

Treatment of CCS presented a challenge in this case due to the patient's co-morbidities. The mainstay treatment of high-dose prednisolone was considered; however, his sarcopenia, proximal myopathy, cardiovascular risk factors, recurrent ascites, and multiple episodes of SBP were compelling relative contraindications to

this therapy. Overall, the risks were thought to outweigh the benefits in his case. Supplemental oral high-protein, high-energy (HPHE) meal replacement drinks were prescribed to correct the patients' sarcopenia and hypoalbuminemia.

4 | CONCLUSION AND RESULTS

Six months after diagnosis, the patient's BMI had remained stable at 31 kg/m² and serum albumin had increased from 27 g/L to 33 g/L. There had been no further episodes of SBP or acute kidney injury (AKI), although recurrent reaccumulation of ascites requiring three weekly large-volume paracenteses remained an issue. The patient's diarrhea had resolved and repeat fecal alpha-one antitrypsin to quantitative hemoglobin ratio was low at ≤ 0.01 . Follow-up colonoscopy at 6 months showed almost complete resolution of the previous polyposis, with only a single 5 mm sessile tubular adenoma in the ascending colon that was resected, as well as 3 diminutive polyps with an inflammatory appearance at the hepatic flexure (Figure 6). Follow-up gastroscopy was unchanged with the antral polyps unchanged in appearance. What defines remission in CCS varies in the literature, but using Pham et al.¹³'s criteria, our patient was considered to be in clinical remission (due to resolution in gastrointestinal symptoms, normalization of biochemistry and improvement in ectodermal abnormalities) and endoscopic remission (with a decrease in polyps to less than 5 isolated polyps remaining without congestion or edema in adjacent mucosa). He also met the criteria for clinical remission using alternate criteria outlined by Hokari et al.³ (score of 1) and partial remission using Watanabe et al.⁴'s criteria (due to the presence of mild endoscopic activity).

Additionally, the decision was made to refrain from anticoagulation therapy for the SMV thrombus as the risk of bleeding was considered too high. Subsequent CT imaging revealed resolution of the SMV thrombus without anticoagulation.

5 | DISCUSSION

CCS is a very rare syndrome which lacks a confirmed etiology and established treatment options. We present what we believe is the first reported case of CCS in a patient with cirrhosis and discuss the diagnosis and treatment challenges this created. In this case the diagnosis was made after a colonoscopy was performed for hematochezia. Prior to presentation with hematochezia the patient's symptoms were primarily related to protein-losing enteropathy, of which CCS is an important but rarely encountered differential

FIGURE 2 Histological features of the removed polyps. (A) Low power view of the gastric antral polyp showing hyperplastic glands within an inflamed fibrovascular stroma. (B) Low power view of a representative colorectal polyp showing elongation and focal dilatation of the crypts and expansion of the stroma. (C) High power view of a colorectal polyp showing a stromal infiltrate of lymphocytes and eosinophils. (D) High power view of a colorectal polyp showing neutrophils in a dilated crypt.

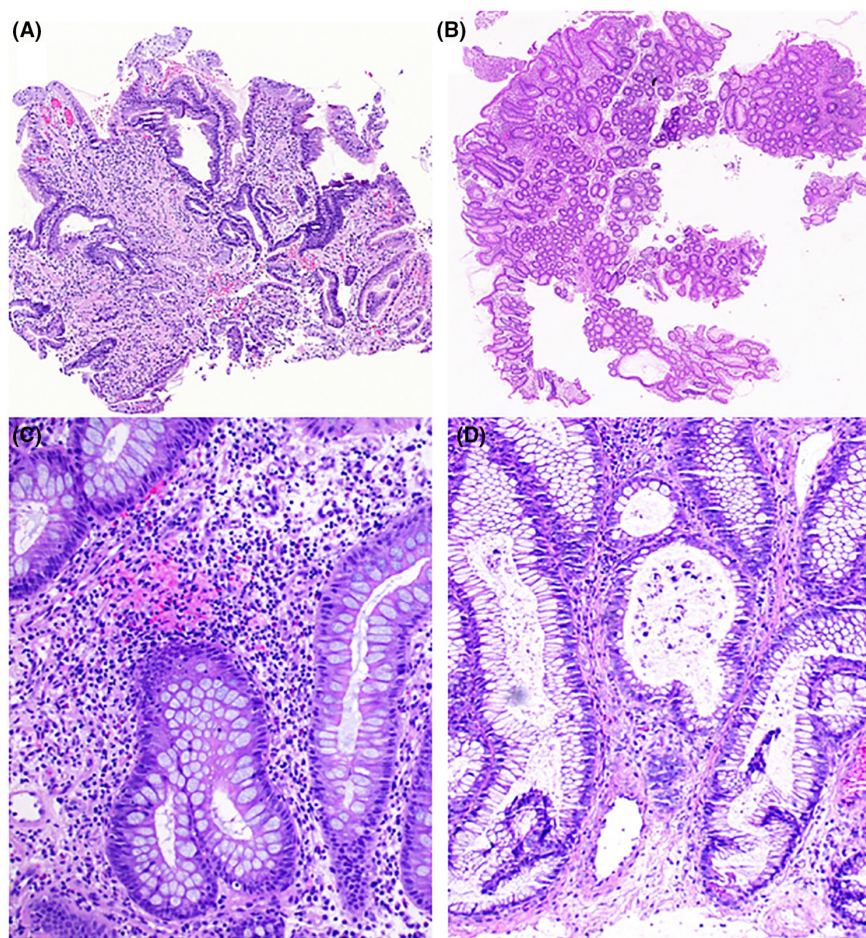


FIGURE 3 Alopecia seen in our patient.



FIGURE 4 Fingernail and toenail onychodystrophy seen in our patient.

diagnosis. Hence, for 6 months preceding the diagnosis of CCS, symptoms and clinical findings were attributed to existing co-morbidities and medications. Although many of the clinical findings could be explained by cirrhosis with portal hypertension, the history of diarrhea was subacute (6 months) and the patient's weight loss and biochemical abnormalities followed a similar time course. Our patient's constellation of diarrhea, hematochezia from intestinal polyposis, alopecia, and nail onychodystrophy raised the

clinical suspicion of CCS, which was later confirmed by histopathological findings.

Histologic findings of increased staining for IgG4 in intestinal polyps found in CCS and treatment responses to immunosuppressants support CCS being the result of immune dysregulation. A potential association between CCS and IgG4-related disease has been raised previously; however, the absence of other autoimmune and

TABLE 3 Additional laboratory findings in our patient.

Laboratory test	Result	Units	Normal range
Fecal alpha-one antitrypsin to quantitative hemoglobin ratio	0.44, 0.36	Ratio	≤0.03
Total globulin	33	g/L	25–45
IgG total level	12.2	g/L	7.0–16.0
IgG1	7.11	g/L	4.05–10.1
IgG2	5.32	g/L	1.69–7.86
IgG3	1.70	g/L	0.11–0.85
IgG4	1.21	g/L	0.03–2.01
Lupus anticoagulant	Positive		
Anti-thrombin III	0.72	U/mL	0.83–1.28
Anti-cardiolipin antibody	8	CU	<20
Anti-beta-2-glycoprotein	0	G units	<20
APC resistance	5.33	Ratio	>1.80
Factor V Leiden	Absent		
Prothrombin 20210G>A	Absent		
Protein S	0.96	U/mL	0.74–1.46
Protein C	0.82	U/mL	0.70–1.30
Ferritin	84	μg/L	45–715
Transferrin saturation	24	%	15–45
Vitamin B12	239	pmol/L	133–680
Serum folate	>51.7	nmol/L	>7.0
25-hydroxy-vitamin D	90	nmol/L	50–150
Anti-nuclear antibody	Negative	titre	
Extractable nuclear antibody	Negative	titre	

IgG4-related diseases in other organs supports that it is a distinct clinical syndrome.¹⁴ Histopathology performed with the broader clinical context provided to the pathologists allowed a diagnosis consistent with CCS to be made and reinforces the importance of multidisciplinary input for early diagnosis observed previously.¹⁴ An unusual feature of the case we present is the absence of diffuse upper gastrointestinal tract involvement. In previous cases at our center, the gastric mucosa has been observed to have diffuse polyposis with similar findings in the duodenum or has shown flattened nodular mucosa. The lack of this diffuse involvement may represent that our patient had a mild disease phenotype.

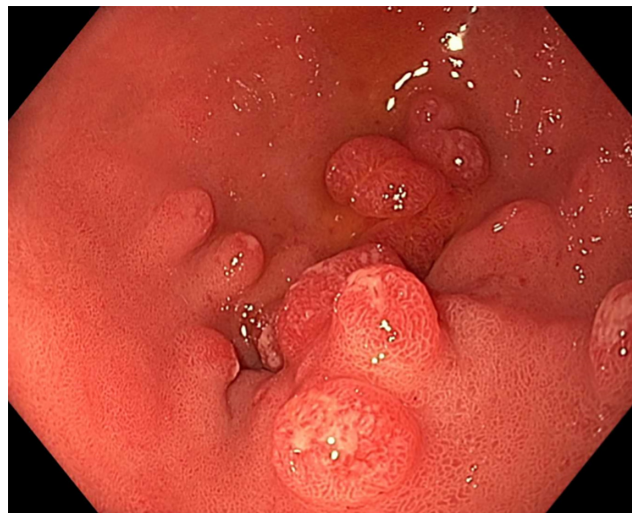


FIGURE 5 Endoscopic images of gastric antral polyps.



FIGURE 6 Endoscopic images of colonic polyps.

Given the hypothesized immune dysregulation in CCS, systemic immunosuppression is considered the most effective treatment.^{3,6,7,15} However, there is still minimal evidence to support whether the treatment has a long-term mortality benefit particularly in patient cohorts at increased risk of side effects from corticosteroids, such as in our case. Patients who have remission induced with corticosteroids usually relapse after completion of induction dosing, meaning an extended wean or long-term low-dose therapy is required.¹⁵ The symptom and mortality benefit would have to be compelling to justify use in these circumstances, and in our case, we felt that the unknown benefit was outweighed by the known risks of corticosteroid therapy. Prior to diagnosis, our patient used regular hydrocortisone at a physiologic dose (equivalent to 5 mg prednisolone) for panhypopituitarism, and at this dose, it was unlikely to have altered the course of his disease. The use of azathioprine and anti-TNFα agents was again felt to be too high-risk in this case due to the history of multiple previous episodes of SBP and an inability to prevent ascites

reaccumulation due to diuretic intolerance. Mesalamine, which acts locally on the colonic mucosa has been shown to be a useful low-risk alternative to corticosteroids in the induction of remission in CCS and may be an appropriate first-line option in similar patient cohorts.^{16,17}

Nutritional therapy, including HPHE supplementation, exclusive enteral nutrition, and total parenteral nutrition, has been used alongside pharmacological therapy to improve the chance of and time to remission in CCS.^{3,4,10} There has also been documented success of induction of remission with nutrition therapy alone. A recent case study by Lipin et al.¹¹ showed induction of remission with a high-protein diet, whereby the patient was prescribed 70 grams of protein per meal in addition to an oral proton pump inhibitor and zinc. Similarly, Russell et al.¹⁰ described a case of CCS that achieved clinical remission after 10 weeks of exclusive enteral nutrition with isocal polymeric formula 1.5L/day (1500kcal/day). A large Japanese retrospective survey of 210 patients with CCS revealed prolonged clinical remission with nutrition therapy alone in one patient having had both total parenteral nutrition and a defined-formula diet.⁴ Due to the possibility of spontaneous remission a causal relationship has not been established between the use of nutritional therapies and induction of remission, although it does appear to be a useful strategy in improving ectodermal manifestations and reducing the direct complications of malnutrition.⁴

Supplemental HPHE enteral nutrition was prescribed in our case due to the significant sarcopenia, weight loss, and hypoalbuminemia the patient had on presentation. The product chosen was a commercially available polymeric meal replacement drink. Sarcopenia is a common complication of cirrhosis with poor long-term outcomes, and HPHE enteral supplementation is routinely used in this patient cohort.¹⁸ Previous studies have shown hypoalbuminemia is an independent risk factor for AKI and recurrent SBP.^{19,20} Our patient's sarcopenia and hypoalbuminemia on presentation were multifactorial with protein losing enteropathy from CCS confirmed by increased fecal alpha-one antitrypsin ratio. Use of nutritional supplements without other specific therapy resulted in clinical remission, with only mild endoscopic activity on follow-up colonoscopy in this case. Serum albumin improved, and there were no further episodes of AKI or SBP during follow-up.^{19,20} Due to the possibility of spontaneous remission which has been reported previously, similar to previous case reports reporting success with nutritional therapy it is possible there is no causal relationship between the use of HPHE nutrition supplements and induction of remission in this case.^{4,10,11,21,22} Subsequently, oral HPHE nutrition supplements have been continued in line with our center's usual management of patients with decompensated cirrhosis rather than for CCS alone.

On presentation, our patient presented with an SMV thrombosis, which presented a treatment challenge due to his bleeding risk. The complication of thromboembolism in CCS has been reported in the literature and is associated with sepsis,²³ malignancy,²⁴ and surgery.²⁵ A case report by Sampson et al.²⁶ also described a patient with CCS who had unprovoked thrombosis that recurred with attempts to stop anti-coagulation after an adequate duration. Interestingly, most of these cases occurred in the setting of an elevated circulating factor VIII and fibrinogen, which has been associated with an increased risk of venous and arterial thrombosis. The mechanism by which this occurs is largely unknown but is thought to occur by the high factor VIII levels enhancing thrombin formation and/or through the induction of acquired activated protein C resistance.²⁷ Our patient had normal fibrinogen and factor VIII levels but a mildly low anti-thrombin III level. He did not meet the criteria for anti-thrombin deficiency as he did not have anti-thrombin activity levels consistently less than 80% of normal. His SMV thrombus was likely a result of his hypercoagulable state associated with his CCS and resolved without intervention. The presence of the thrombus within the SMV as a complication of portal hypertension alone would be unusual as there was no extension into the portal vein. In cases of CCS with significant bleeding risk and uncomplicated SMV thrombosis, it appears reasonable to monitor with short-interval follow-up imaging.

In summary, we describe a case of CCS in a patient with decompensated cirrhosis who had successful induction of remission with nutritional supplementation alone. CCS is an important but rarely encountered cause of hematochezia and protein-losing enteropathy, and this case reinforces the importance of early endoscopic assessment and multidisciplinary input where intestinal polyposis is seen. We propose that early institution of HPHE enteral supplementation as nutritional therapy should be offered to all patients due to its' low risk profile and role in limiting the complications from protein-losing enteropathy. This is particularly salient in cases where there are compelling contraindications to the use of high-dose corticosteroids for induction of remission due to the possibility of induction of remission from nutritional therapy alone and spontaneous remission. The addition of a steroid-sparing agent can also be considered on an individual basis, although evidence is lacking.

AUTHOR CONTRIBUTIONS

Anthony J. Deacon: Conceptualization; writing – original draft; writing – review and editing. **Paris Hoey:** Writing – original draft; writing – review and editing. **Florian Grimpén:** Writing – review and editing. **Leo Francis:** Investigation; writing – original draft; writing

– review and editing. **Ian Brown:** Investigation; writing – original draft; writing – review and editing. **Barbara Leggett:** Writing – review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare that there is no conflict of interest.

DATA AVAILABILITY STATEMENT

Information used for this case report can be made available from the corresponding author upon request.

ETHICS STATEMENT

This study did not include experiments on animals or humans. The patient consented to the use of their personal data for the purpose of this case report.

CONSENT

Written informed consent was obtained from the patient to publish this report in accordance with the journal's patient consent policy.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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